

QMHP-CoPro

Heavy-Fluxonium Erasure-Conversion Coprocessor Architecture

Version 1.5.8f — Full Standalone Technical Whitepaper

Matched-Construction Branch Screen · Free-Location Bracket · Realistic-Location Cost Study ·

Consolidated Verification Register

24 August 2026

DOCUMENT STATUS

TRL-2 architecture and falsification dossier. No experimental validation of QMHP-CoPro is claimed. v1.5.8f is the full standalone edition of the programme record as of 24 August 2026: it carries the complete v1.5.8e technical content with edit register E1–E20 applied inline, integrates the AMD-C realistic-location cost study (T.12') and distance-trend statistic (T.13'), and prints the consolidated three-tier verification register produced by the August 2026 external review cycle. The branch-screen conclusion is NO SEPARATION ESTABLISHED between Branch A and the conventional control under matched construction; the v1.5.8d adverse screen remains WITHDRAWN as a construction artifact. All logical results are MODEL-DEPENDENT screens; measured branch-specific hardware ledgers and a full gate-propagation circuit model remain required before fabrication selection.

Scope of this edition

This edition carries forward the still-valid v1.5.6a Hamiltonian, coherence decomposition, error ledger, validation thresholds and stop rules as rewritten current sections, with every supersession stated explicitly. It retains Branches A/B as erasure-conversion research branches and Branch C as the conventional-fluxonium null control, evaluated at equal wall clock with the logical-hazard metric λ_L . It retains the non-perturbative dressed logical-blind null and the full fabrication, finite-photon, Floquet, pointer-dephasing, IQ-discrimination and Purcell analyses, updated with truncation labels (E13, E18). It re-executes the row-by-row phenomenological QEC-MAP in an independent clean-room implementation after removing four one-directional construction asymmetries, adds the ideal free-location bracket, the location-efficiency sweep and break-even f^* , and — new in this edition — the cost-charged realistic-location study T.12' with the P6-E6' joint acceptance contract, the T.13' distance-trend statistic under the EXPLORATORY label, and the executable channel-convention table. Prior revisions are provenance only; this document is the current specification.

Document control and precedence

Field	Current disposition
Version	1.5.8f — full standalone technical whitepaper, edit register E1-E20 applied
Date	24 August 2026
Technology readiness	TRL-2: architecture, computation and falsification programme; no device-level experimental validation claimed
Preserved baseline	QMHP-CoPro v1.5.6a, 25 July 2026; intermediate patch v1.5.7, 19 August 2026
Superseded editions	v1.5.8d (screen WITHDRAWN); v1.5.8e both builds (operative content carried here); AMD-C r1-r6 and ERRATA-1 (change-record artifacts); Master condensed edition (executive companion)
Provenance anchors (E17)	OPERATIVE archival base: v1.5.8e 117-page build, SHA-256 d334cda822c07c212be1fac850a34debe5ea66ee45899e56b44e19154f86c6c. PROVENANCE ONLY: 109-page render, 83aeb3a2d8ebcb8f627989557350d9c172ad5bc20c3edc561b8a5e091cc4b658. Complete S1 shot grids and executable listings S5.1-S5.6 are carried by reference to the operative anchor and re-shipped in the release bundle
Branch selection	NO SEPARATION ESTABLISHED between Branch A and Branch C under matched construction. Neither an adverse nor a favourable screen is decision-bearing until both arms are charged on the same basis with measured inputs
External literature cutoff	21 August 2026

v1.5.8f REVISION NOTICE

v1.5.8e audited and re-executed the decision machinery first run in v1.5.8d. Four features of the v1.5.8d construction ran one way only: Branch A was charged at its P4 acceptance ceiling while Branch C used a published record; Branch A's located load was charged as a uniform Pauli while Branch C's physically analogous T1 decay was charged as twirled amplitude damping; Branch C's ledger omitted approximately $1.95\text{e-}3$ /round of mechanism classes ($1.15\text{e-}3$ additions plus the $0.8\text{e-}3$ correlated-reserve uplift) that its own contract requires; and no free-location bracket was run. Correcting all four gives $\lambda_A/\lambda_C = 0.94$ [0.88, 1.01] at $d=3$, 0.92 [0.79, 1.08] at $d=5$ and 1.15 [0.97, 1.36] at $d=7$ — no separation at any distance. With the ideal free-location bracket the ratios are 0.43 [0.39, 0.47], 0.20 [0.17, 0.24] and 0.15 [0.13, 0.18]. The v1.5.8d statement that Branch A does not presently earn its extra hardware is WITHDRAWN as a construction artifact. v1.5.8f adds the cost side: T.12' charges check-induced logical error, check time and finite location efficiency under the release's own contracts, finds the decoder-level break-even at $1.1\text{--}1.3\text{e-}2$ per data qubit per round, and finds the programme ledger ceiling binding $\approx 7.5\times$ earlier — producing the P6-E6' joint contract and the rejection of any $5.0\text{e-}3$ acceptance bar. The August 2026 external review cycle (five independent referees plus execution-level verification) is consolidated in §15; every defect found on any side is registered in §16 and closed.

Decision brief (E9)

Current state.

Continue as a gated research programme. Under the matched-construction row-wise map, Branch A and Branch C are statistically indistinguishable at every tested distance with no erasure-location credit; Branch A is lower at $\geq 95\%$ confidence at every distance under the ideal free-location bracket. Because Branch-A inputs are allocations and Branch C is a cross-device prior, neither branch is selected and neither is excluded. The next decision-bearing numbers are hardware numbers.

Excluded claims — this edition does NOT support.

Any processor-performance or fault-tolerance claim; any absolute p_L as a hardware prediction (QEC-MAP-PHENO is a model screen); the withdrawn v1.5.8d conclusion, and equally any claim that Branch A has demonstrated advantage — the zero-location margin traces to lower charged load per unit time, not erasure conversion (premise-kill watch, §13.2); any confirmatory distance-trend claim from T.13' (EXPLORATORY); exact finite-sample ensemble counts as reproducibility targets (E19) — the reported object is the rejection rate, $\approx 6.2\text{--}6.7\%$ across independent seed blocks.

Top blocker	Status / evidence	Owner	Ordering gate (E10)
P6-E7 leakage-to-syndrome coupling	BLOCKING. Two LRU1 implementations differ $3.3\times$ at $d=3$ ($8.52e-3$ vs $2.840e-2$) and reverse the distance trend	B. Duncan (programme lead)	Adjudication rule and protocol frozen before comparative decoder data are opened
Driven sink survival $T_{\text{sink}(n)}$	BEYOND-DEMONSTRATED: 3–6 ms class vs $\approx 20\ \mu\text{s}$ An-like during-check comparator	B. Duncan	P6-E3 first cooldown; $q_{\text{miss}} \leq 5e-4$ re-optimised jointly with the cost-side gate (E3)
Full gate propagation (Stim-level)	Required before any absolute p_L is physical; tooling unblocked	B. Duncan	Frozen decoder/noise model before holdout (P7-pre)
Joint-device Branch-C control	Branch C is a cross-device literature prior	B. Duncan	Same-device, same-cooldown measured control before fabrication commit (§8.3)
Coupler/readout spectral replan	Readout moved to $\approx 4.30\ \text{GHz}$; legacy coupler plan not revalidated	B. Duncan	Regenerated spectral plan before tile-scale fabrication (§11)
Purcell filter $\kappa(\omega)$	34–36 dB stopband at dressed 3.395056 GHz mandatory under current anchors	B. Duncan	Measured $\kappa(\omega)$ around the 4.302 / 3.3951 GHz bands (P6-E2)

Notation quick reference.

Term	Meaning
Branch A / B / C	Half-flux heavy fluxonium ($ 0\rangle/ 2\rangle$ logical, $ 1\rangle$ monitored sink) / integer-flux erasure fluxonium / conventional-fluxonium null control
Pe, Pu	Located-erasure load per round ($1.543e-2$) / unlocated ledger per round (baseline $8.50e-3$; programme ceiling $1.0e-2$; headroom $1.50e-3$)
λ_L , HR	Logical hazard $-\ln(1-p_{L,\text{round}})/t_{\text{round}}$; hazard ratio λ_A/λ_C at equal wall clock ($t_A = 20\ \mu\text{s}$; $t_C = 13.5\ \mu\text{s}$ prior, 12–15 sensitivity)
f, f*, p_induced, t_chk	Location efficiency; break-even f vs C-as-published (0.524/0.598 at $d=3/5$); check-induced logical error per data qubit per round; added check time
twirlAD	Pauli-twirled amplitude damping, $p_X = p_Y = p_Z = \gamma/4$ (executed convention on both branches)
δ (E14)	Junction phase $\delta \equiv \varphi - \varphi_{\text{ext}}$; at $\varphi_{\text{ext}} = \pi$, $\sin(\delta/2) = -\cos(\varphi/2)$, parity-EVEN
F6 / F8 / F11	Pointer-dephasing functional; Purcell weight $\Gamma_P = \kappa(\omega_{\text{if}}) \langle f a i\rangle ^2$; ledger-conserving row-wise mapping rule
Tier A / B / C	Release execution / reconstruction-level re-execution / independent replication from stated equations (§15)
Status labels	VERIFIED-COMPUTATIONAL, MODEL-DEPENDENT, CONSTRUCTION-SENSITIVE, HARDWARE-BOUND, BEYOND-DEMONSTRATED, EXPLORATORY, WITHDRAWN — removing a label changes the meaning of the statement

1. Executive decision summary

Branch A remains physically interesting for one reason: the half-flux $\{|0\rangle, |2\rangle\}$ encoding is symmetry-protected against the first-order charge transition while the dominant decay $|2\rangle \rightarrow |1\rangle$ converts to a located event on a monitorable sink. The same charge-parity selection rule that provides the protection forbids the direct logical charge drive (hence Raman control) and leaves an unlocated $|2\rangle \rightarrow |0\rangle$ quasiparticle bypass as the central unbudgeted threat (§3.4). Branch B carries direct 2026 single-fluxonium erasure-conversion hardware evidence at integer flux; the same literature also shows measured check back-action can dominate the benefit, which is why the An-like comparator enters this edition as an adverse cost anchor rather than a design constant. Branch C is the discipline: a conventional fluxonium charged on its own published record, contract-completed for matched comparison.

The programme question is narrow: under an explicit row-wise error map, matched charging conventions and equal elapsed time, does an erasure-conversion branch beat the conventional control after all branch-specific costs are charged? v1.5.8e executed the first matched model screen and returned NO SEPARATION. v1.5.8f charges the cost side of the location benefit (T.12') and finds the acceptance question turns on check-induced logical error under a binding ledger ceiling. Neither result substitutes for hardware; both sharpen exactly what hardware must measure.

1.1 What has been computationally closed

Domain	Frozen result (status)
Half-flux spectrum / parity	f01 = 0.175710013 GHz, f02 = 3.581724312 GHz; $ n_{02} $ at numerical floor; $ \langle 0 \sin(\delta/2) 2\rangle = 0.295705232$ with δ per E14. INDEPENDENTLY REPLICATED to all nine figures by two external builds
Quasiparticle bypass (H.5)	$\Gamma_{qp}/x_{qp} = 4.106e10\text{ s}^{-1}$ ($\Delta = 180\text{--}210\text{ }\mu\text{eV}$ band: 3.945–4.261e10); $x_{qp} \leq 1.218e-10$ required at the 5 s^{-1} P1 bound — $\approx 4.6\times$ below the best published resonator value. BEYOND-DEMONSTRATED
Readout operating point	Eq.(7) seed 4.314628047 GHz; exact dressed null 4.301974466 GHz at production truncation (10,12), register spot-check value 4.301975383 GHz at (14,25), $\Delta = 0.92\text{ kHz}$ inside the stated band (E13). Pulls $-9.507/+8.721\text{ MHz}$; contrast 18.228 MHz; sink line 4.310696 GHz; dressed f12 = 3.395056 GHz, truncation-stable
Fabrication / collision gate	400/400 $\sigma=2\%$ samples retain a dressed null at the declared seed; rejection rate $\approx 6.2\text{--}6.7\%$ pooled across two independent 20-seed blocks; $32/400 = 8.0\%$ is a $+1.1\sigma/+1.5\sigma$ single draw (E19)
Static Purcell channel (E18)	F8-exact weight 0.01182198 at (12,18)/(14,25), 0.01182201 at production (10,12); bare-admixture overlap 0.011815 is a distinct object; unfiltered $ 2\rangle \rightarrow 1\rangle$ lifetime $\approx 6.73/4.49\text{ }\mu\text{s}$ at $\kappa/2\pi = 2/3\text{ MHz}$; 34–36 dB stopband at 3.395056 GHz mandatory
Pointer / Floquet / IQ	F6 pointer functional supersedes the n^3 proxy (784 $\times$ overestimate; 0.021–0.058 photon ceiling WITHDRAWN); $ 1\rangle+9\gamma \rightarrow 14\rangle$ line bounded inert in $n = 5\text{--}8$; worst ideal-Gaussian IQ point SNR 6.91, assignment error $5.10e-7$ ($\kappa/2\pi = 2\text{ MHz}$, $n \approx 5$, $T = 1.5\text{ }\mu\text{s}$, $\eta = 0.3$)
Row-wise map (F11)	$p_{opp} = n_{data}/\text{alloc}/(n_{opp}q_{per_event})$; all fourteen $d=3$ rows reproduce to $<1e-9$; ledger re-aggregates to $8.500e-3$ exactly. INDEPENDENTLY REPLICATED
Matched branch screen	NO SEPARATION at $d = 3/5/7$ without location credit; A lower at $\geq 95\%$ at every distance with the ideal bracket; v1.5.8d CONSTRUCTION-SENSITIVE and WITHDRAWN
Cost-charged study (T.12')	Decoder break-even $p^*_{induced} = 1.1\text{--}1.3e-2$; ledger ceiling binds at $1.5e-3$ headroom ($\approx 3.3e-4$ after An-like sink conversion); P6-E6' joint contract registered; $5.0e-3$ bar REJECTED
Filter sharpness	Synthetic stopband 3.2056–3.6727 GHz, maximum required depth 37.65 dB; 464.24 MHz transition to the lowest readout root ($\geq 81.1\text{ dB/GHz}$); paired same-device separation $\geq 761\text{ MHz}$, median 905 MHz

1.2 Open and hardware-bound

Filtered, driven sink lifetime $T_{sink}(n)$ under the actual P6-E tone including sink destruction and destination-resolved leakage; MIST, TLS and QP/radiation response under the delivered spectrum; array, package and coupler modes; measured $\kappa(\omega)$ around 4.302 and 3.3951 GHz; P0d/P1 branch-resolved rates on a single device and cooldown with reset ON/OFF; the coupler-band redesign and EOL mapping; the Branch-B executable ledger and round duration; the mechanism-resolved erasure-use study with measured (f , $p_{induced}$, t_{chk}) per P6-E6'; and the full circuit-level gate propagation model. Every item carries a named owner and an event-anchored ordering gate in the decision brief.

2. Physical architecture and branch definitions

Branch A is a half-flux heavy fluxonium with $(E_C, E_J, E_L)/h = (0.60, 5.72, 1.58)\text{ GHz}$ at $\phi_{ext} = \pi$, logical states $\{|0\rangle, |2\rangle\}$ and a monitored $|1\rangle$ sink. Charge-parity selection forbids the first-order $|0\rangle \leftrightarrow |2\rangle$ charge transition ($|n_{02}|$ at the numerical floor), while $|2\rangle \rightarrow |1\rangle$ is charge-allowed with $|n_{12}| = 0.659915$ and constitutes the located-erasure conversion channel; uniform Pauli placement of that load would put 4/3 the per-axis weight of the executed twirled-amplitude-damping surrogate (E12). Branch B is an integer-flux erasure fluxonium with 2026 external hardware evidence and its own contract (Appendix M, §8.2). Branch C is the conventional-fluxonium null control: a cross-device literature prior with a six-row published ledger (E11), completed against its own §§2.3/8.3 contract for matched comparison. Common doctrine: no branch is charged on another's convention; every comparative screen must satisfy the §8.4 matched-construction conditions or

be labelled CONSTRUCTION-SENSITIVE and excluded from decisions.

3. Branch-A device physics and erasure mechanism

3.1 Hamiltonian and nominal spectrum

$H/h = 4E_C n^2 + \frac{1}{2}E_L \varphi^2 - E_J \cos(\varphi - \varphi_{\text{ext}})$, $[\varphi, n] = i$, at the nominal point $(E_C, E_J, E_L)/h = (0.60, 5.72, 1.58)$ GHz, $\varphi_{\text{ext}} = \pi$. The baseline phase-grid method uses 3201 uniform points on $[-8\pi, 8\pi]$ with a second-difference kinetic operator; eigenvectors are normalised in the discrete measure and operator matrix elements are evaluated in the converged eigenbasis.

Quantity	Value	Status / note
f01	0.175710013 GHz	VERIFIED-COMPUTATIONAL; well-pair splitting
f02	3.581724312 GHz	VERIFIED-COMPUTATIONAL; logical gap
f03 / f04 / f05	5.207305246 / 7.982111939 / 10.815572 GHz	VERIFIED-COMPUTATIONAL
bare ω_{12}	3.406014 GHz	= f02 – f01; erasure-emission seed
$ n01\rangle / n03\rangle / n12\rangle / n23\rangle / n25\rangle$	0.080035 / 0.591046 / 0.659915 / 0.581166 / 0.418627	charge matrix elements, six figures, independently replicated
$ n02\rangle$	$\approx 1e-12$	charge-parity zero; the protection mechanism
$ \langle 0 \sin(\delta/2) 2\rangle $	0.295705232	$\delta \equiv \varphi - \varphi_{\text{ext}}$ (E14): $\sin(\delta/2) = -\cos(\varphi/2)$, parity-EVEN; the loop-phase operator $\sin(\varphi/2)$ is parity-odd and vanishes

3.2 Numerical method and convergence discipline (E16 applied)

Quantities feeding root-finds, cancellations or Richardson extrapolations are retained at higher internal precision than display precision. The H.4 curvature regression uses phase steps of 0.020, 0.010, 0.005 and 0.0025 — RADIANS of φ_{ext} , converted to GHz/ Φ_0^2 by $(2\pi)^2$ — with raw second derivatives 4426.524849, 4873.297283, 5014.089593 and 5051.757635 GHz/ Φ_0^2 , extrapolating to 5064.55 GHz/ Φ_0^2 under full h^2 -elimination (a linear-in- h^2 fit lands ≈ 5050 and is not the registered scheme). Reading the steps as Φ_0 fractions reproduces an external referee's low raws exactly ($0.0025 \Phi_0 \rightarrow 4635$) — the caption exists so that misread cannot recur. The large curvature is not itself a first-order sweet-spot failure ($df02/d\Phi$ vanishes at half flux) but makes flux excursions important for coherence and for readout-band parity breaking, since n_{24} grows linearly away from symmetry at $\approx 8.66/\Phi_0$ — the origin of the 13 MHz collision clearance: $|\Delta_{24}|_{\text{min}} = g \cdot 8.66 \cdot (1e-4)/\sqrt{(1e-4)} = 12.99$ MHz for the declared $\delta\Phi \leq 1e-4 \Phi_0$ excursion and $p_{\text{mix}} \leq 1e-4$ allowance. Dressed-root convergence: the production sweep used $N_q=10/N_{ph}=12$ only after spot checks against $N_q=14/N_{ph}=25$ showed 0.748–1.202 kHz root deviations, negligible against the 13 MHz screen; printed nine-figure register values carry their truncation label (E13, E18).

3.3 Erasure mechanism

Desired erasure is $|2\rangle \rightarrow |1\rangle$ conversion into the monitored sink, entering the ledger as the located load $P_e = 1.543e-2$ per data qubit per 20 μs round. The sink is interrogated by the logical-blind readout of §6; sink residence, destruction and back-action are the P6-E measurement programme. Unconditional sink reset each round is the baseline cadence; the R.5 register shows reset cadence changes temporal support only, not a p_L multiplier, under the Pauli-equivalence contract — persistent residence is the separately-screened LRU variant of §14 (T.9).

3.4 Quasiparticle bypass and the P1 channel bound (H.5)

$\Gamma_{20}^{\wedge} q_p / x_{qp} = 16 (E_J/h) |\langle 0|\sin(\delta/2)|2\rangle|^2 \sqrt{(2\Delta/(h f_{02}))}$. With $\Delta = 195 \mu\text{eV}$, $f_{02} = 3.581724$ GHz and the replicated matrix element, $\Gamma_{qp}/x_{qp} = 4.106e10 \text{ s}^{-1}$; in the small-probability limit $P_{20}^{\wedge} q_p/x_{qp} = 8.213e5$ per 20 μs ; the $\Delta = 180\text{--}210 \mu\text{eV}$ band spans $3.945\text{--}4.261e10 \text{ s}^{-1}$. Holding the P1 allocation of 5 s^{-1} therefore requires $x_{qp} \leq 5/(4.106e10) = 1.218e-10$ — about $4.6\times$ below the best published resonator-mode value ($5.6e-10$). This is the central BEYOND-DEMONSTRATED demand of the architecture and the reason P0d/P1 on one vehicle and cooldown, reset ON/OFF, is the first conjunctive hardware gate. The half-flux sine channel being large is not a modelling artifact: it is the canonical quasiparticle physics of fluxonium at the symmetry point, and it is what the sink conversion must outrun.

4. Control architecture

Direct logical charge drive is parity-forbidden; single-qubit control is Raman-mediated through the allowed transitions. P2 targets: cycle error $\leq 1e-4$, leakage $\leq 2e-4$ per operation, with no control-induced rise in the QP or thermal registers. The schedule assumption $N_{1Q} \leq 5$ operations per round is a provisional ZERO-MARGIN assumption and is flagged as such: with Raman control replacing the withdrawn direct drive, five operations per 20 μs round is optimistic, and a compiled-schedule artifact is the registered stress test (§16

recommendations, F5). Record-edge comparators (FTF CZ 99.922%; direct fluxonium CNOT 99.94%) are context for the P4 ceiling, not transferred performance.

5. Reset and coherence contracts

Reset is represented as a complete map R , not a lifetime scalar. P0d reports residual sink population $r_{\text{sink}} = \text{Tr}[|1\rangle\langle 1| R(|1\rangle\langle 1|)]$, the logical-subspace process error of $\Pi_L R(\cdot) \Pi_L$ relative to identity, leakage/outflow from $|2\rangle$, conditional phase, return-ramp back-population and any QP generation. Targets: $t_{\text{reset}} \leq 8 \mu\text{s}$, residual $|1\rangle \leq 1\%$, logical back-action $\leq 2.5e-4$, thermal sink lifetime $\tau_{\text{th}} \geq 1.8 \text{ ms}$. The joint target implies a switchable damping ratio $\geq 225\times$ on the same 176 MHz sink transition while retaining the back-action bound — the engineering requirement stated plainly. Thermal floor: $p_{1,\text{eq}} = 1/[1 + \exp(\hbar \omega_{01}/k_B T)] = 0.363/0.396$ at 15/20 mK, giving 0.401–0.438% sink loading per 20 μs round at $\tau_{\text{th}} = 1.8 \text{ ms}$ — the origin of the thermal_unflagged ledger row. Coherence is a decomposed object (P3): $[1 - W_\varphi(t_{\text{round}})]/2 \leq 1.8e-3$, equivalent to $T_\varphi \geq 5.55 \text{ ms}$ at the 20 μs exponential special case, reported branch-resolved so that averages cannot hide below-threshold windows.

6. Readout architecture: P6-S and P6-E

6.1 Role split

P6-S is syndrome and end-of-line readout: three-state discrimination with a complete confusion matrix, delivered-pulse MIST/Purcell/higher-mode characterisation, and amplifier/latency/calibration closure. P6-E is the mid-circuit data-qubit erasure interrogation: logical blindness between the computational states while resolving the sink, dynamic miss decomposition, sink survival, check-induced logical error and leakage, and the decoder net-benefit gate. The $d=3$ baseline sets the count of mid-circuit P6-E checks to ZERO; the check-back-action row is exactly zero in the baseline ledger and is not a free budget for another mechanism. P6-E remains a research branch and hardware qualification gate for possible reinstatement under the P6-E6' contract of §10.1: the ideal bracket shows the value of location is large enough to be worth paying for, so the measured cost of obtaining it is the decisive unknown rather than a deferred one.

6.2 Dispersive diagnostic and sign convention

For $H = \sum \omega_{ij}|i\rangle\langle j| + \omega_r a^\dagger a + g n(a + a^\dagger)$, the Eq.(7) χ convention is the NEGATIVE of the physical cavity pull $E(i,1) - E(i,0) - \omega_r$. At $\omega_r = 6.5 \text{ GHz}$ and $g/2\pi = 150 \text{ MHz}$ the ten-level Eq.(7) values are $(\chi_0, \chi_1, \chi_2) = (-5.366823, -0.305447, +7.214942) \text{ MHz}$. The sign does not alter $\chi_0 = \chi_2$ roots or separability magnitudes but does set drive detuning, Stark direction and MIST asymmetry; physical pulls are used for those.

6.3 Exact dressed logical-blind null (E13 applied)

The operating point is the non-perturbative dressed null $\delta_0 = \delta_2$ of the coupled system with adiabatic labelling; Eq.(7) supplies the search seed only (4.314628047 GHz; correction -12.653 MHz). Production truncation $N_q=10/N_{ph}=12$ gives $f_r = 4.301974466 \text{ GHz}$ — two fully independent implementations agree to nine figures; the previously printed register value 4.301975383 GHz is the $N_q=14/N_{ph}=25$ spot-check, $+0.92 \text{ kHz}$, inside the stated band. At the null: physical logical pulls $\delta_0 = \delta_2 = -9.507 \text{ MHz}$, sink pull $\delta_1 = +8.721 \text{ MHz}$, sink/logical contrast 18.228 MHz, sink-line drive 4.310696 GHz, dressed emission $f_{12} = 3.395056 \text{ GHz}$ (truncation-stable).

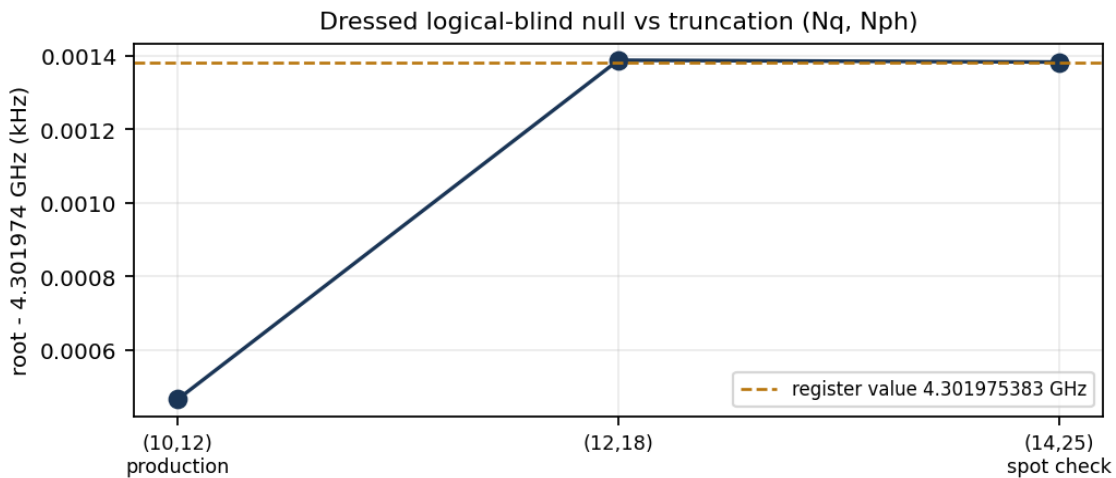


Figure 1. Dressed null vs truncation (regenerates via build_master_figures.py from the shipped replication script).

6.4 Fabrication ensemble and collision gate (Appendix C content; E19 applied)

The 400-device ensemble perturbs (EJ, EC, EL) independently at $\sigma = 2\%$ (declared seed 20260725). It is a controlled static study, not a fab-yield model: resonator disorder, coupler variation, process covariance, package modes, junction-array modes and flux-line calibration covariance are outside the sample. Every sample retains a dressed null (400/400); the collision gate rejects $|\omega_{24} - f_r| < 13$ MHz per device against its own exact dressed root — mandatory device-level accept/reject basis, with the perturbative root retained for diagnostics only (near-equal aggregate rejection is accidental with respect to individual devices: 16 PT-only fails, 14 dressed-only, 18 both). Percentiles carry bootstrap intervals: dressed-root p5 4.19697 [4.17721, 4.21222], median 4.30492 [4.29698, 4.31423], p95 4.41397 [4.40199, 4.42374] GHz; dressed f12 p5/median/p95 3.26983/3.40122/3.54121 GHz; sink-contrast min/p5/median/p95 14.399/16.018/18.237/20.551 MHz. E19 reporting standard: the normative object is the rejection RATE — $\approx 6.2\text{--}6.7\%$ pooled across two independent 20-seed blocks (6.66% [6.12, 7.21] and 6.19% [5.66, 6.72], seeds 1-20 shipped), dispersion consistent with Binomial(400, p); the release's $32/400 = 8.0\%$ is a $+1.1\sigma/+1.5\sigma$ single draw. Exact finite-sample counts are adjudicable only with a declared stream→parameter draw order.

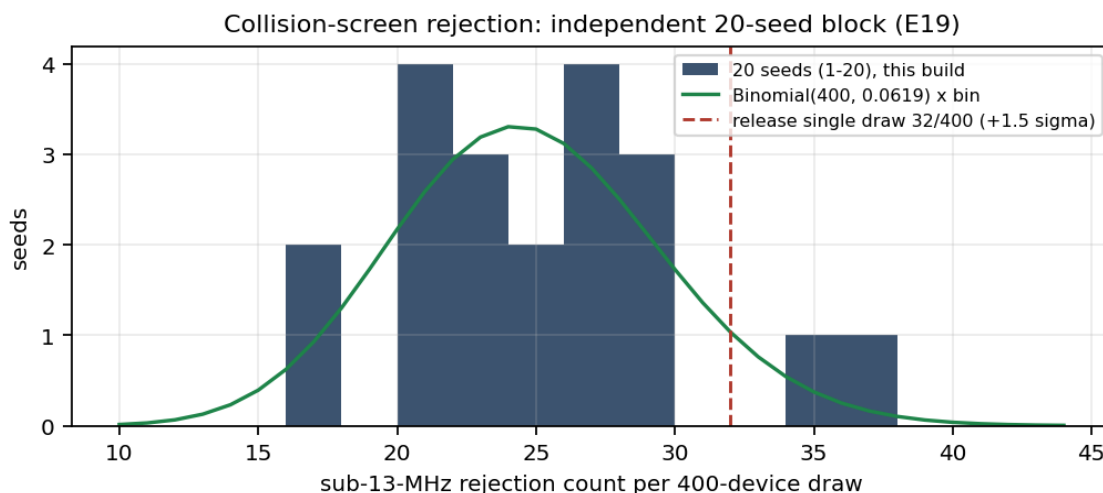


Figure 2. Independent 20-seed collision-screen block (ensemble_20seed.json, shipped).

6.5 Mode census and wiring

The tile-scale plan must include 17 per-device readout roots with common and local filtering; the 17-device spectral-allocation stress test needs a median of 3 and p95 of 5 feedlines for ≥ 10 MHz same-feedline separations — an OPEN hardware/EM requirement charged to the cost-reclosure test (kill criterion 3), not yet executed.

7. P6E-DRIVEN campaign registers

Stage 1 — transient dark loading: the linear-cavity transient trajectory is exact within its model; M1 is the exact dark-loading multiplier; square pulses overshoot the dark steady state at narrow linewidth, and the registered 100 ns cosine-edged pulse family removes most of the overshoot in the 1-5 MHz range. Stage 2a — pointer dephasing: the registered gate is the F6 functional $D = (\kappa/2) \int |\alpha_0(t) - \alpha_2(t)|^2 dt$, coherence $\approx e^{(-D)}$; at matched logical occupation $n_L = 0.02$ and $\kappa/2\pi = 1$ MHz the superseded far-detuned mapping overestimates per-check error by 784 $\times$, and the previously quoted 0.021-0.058 photon ceiling is WITHDRAWN as a wrong-regime bound. Stage 3 — Floquet census: the leading higher-level target $|1\rangle + 9$ photons $\rightarrow |14\rangle$ at the sink-line drive is found and bounded inert in $n = 5\text{--}8$. Stage 4 — ideal IQ discrimination: the worst executed ideal-Gaussian point ($\kappa/2\pi = 2$ MHz, $n \approx 4.97$, $T = 1.5$ μ s, $\eta = 0.3$) gives SNR = 6.91 and assignment error $5.10e-7$, clearing the $5e-4$ target with the worst individual logical pointer, not the average. Stage 5 — exact Purcell channel (E18): the executable definition is F8, $\Gamma_P = \kappa(\omega_{if})|f_a|_i|^2$; the exact dressed weight is 0.01182198 at (12,18)/(14,25) and 0.01182201 at production (10,12); the bare-admixture overlap 0.011815 is a related but different object; the perturbative estimate $(g n_{12}/(\omega_{12} - f_r))^2 \approx 0.0122$ is $\approx 3\%$ high. Unfiltered $|2\rangle \rightarrow |1\rangle$ lifetime is $\approx 6.73/4.49$ μ s at $\kappa/2\pi = 2/3$ MHz; the 1.8 ms anchor demands 267 $\times$ /401 $\times$ suppression, $\times 10$ allocation $\Rightarrow$ the mandatory 34-36 dB stopband at 3.395056 GHz. Stage 6 — sink-branch composition and the driven-sink survival arithmetic feed the T_{sink} requirement of §13.

8. Branch contracts

8.1 Branch-A unlocated ledger (acceptance accounting contract)

The $8.50e-3$ /round unlocated ledger is an acceptance accounting contract — allocations the hardware must beat, not a performance projection. Rows, per data qubit per 20 μ s round unless classed otherwise:

Mechanism	Allocation	Opportunity class	Channel
two qubit CZ	3.200e-03	inc	depol2end
pure dephasing	1.800e-03	data	Z
one qubit aggregate	5.000e-04	gate1q	depol1
measurement decision	4.000e-04	meas	meas
missed erasures	2.500e-04	data	depol1
QP bypass	1.000e-04	data	ampXY
thermal unflagged	1.000e-04	data	ampXY
residual ZZ	2.500e-04	data	Z
nonadiabatic control	2.000e-04	data	depol1
reset backaction	2.500e-04	data	depol1
readout induced	1.000e-04	data	depol1
coupler higher leakage	1.500e-04	data	depol1
correlated reserve	1.000e-03	pair	paircorr
other residual	2.000e-04	data	depol1
TOTAL (re-aggregation exact)	8.500e-03	—	—
Located erasure load Pe	1.543e-02	data	twirlAD × location efficiency f

Section-9 SCALING box (applied literally in T.12’): time-linear, time/burst-, flag- or control-dependent and dephasing-envelope rows (pure_dephasing, missed_erasures, QP_bypass, thermal_unflagged, residual_ZZ, correlated_reserve) and Pe itself rescale with round duration; fixed rows carry at parity.

8.2 Branch B (Appendix M): seeds and evidence

Liu et al. (arXiv:2601.11086, 2026): zero-flux single-fluxonium 0–2 erasure conversion, seed EC/EJ/EL $\approx$ 1.72/7.07/0.32 GHz; $f_{01} \approx$ 5.77 GHz; $>4\times$ postselected lifetime (193→869 μ s); inferred $\approx 7.2e-5$ measurement-induced logical dephasing per check — the favourable cost comparator. An et al. (arXiv:2607.27123, 2026): g–f logical encoding with e sink, seed 1.392/5.056/0.193 GHz; $8.4\times$ postselected f-state lifetime; ancilla-free mid-circuit erasure checks with $7.8 \pm 1.7\%$ false negatives and $2.49 \pm 0.19\%$ added error per check; e→g lifetime collapses to $\approx 20 \mu$ s during the check — the adverse cost comparator. These are parameter-search seeds and cost anchors; performance is not copied. Branch B’s executable ledger and round duration remain open (T.14).

8.3 Branch C contract

Row (published six-row prior)	Value	Channel
T1 amplitude damping	1.1187e-02	twirlAD
pure dephasing	2.9974e-03	Z
one qubit gates	1.8000e-04	depol1
two qubit CNOT	1.6000e-03	depol2end
syndrome measurement	8.8889e-04	meas
correlated reserve	2.0000e-04	paircorr
SUM (as published)	1.7053e-02	—
+ residual ZZ (contract addition)	2.5000e-04	depol1
+ nonadiabatic control (contract addition)	2.0000e-04	depol1
+ reset backaction (contract addition)	2.5000e-04	depol1
+ readout induced (contract addition)	1.0000e-04	depol1
+ coupler higher leakage (contract addition)	1.5000e-04	depol1
+ other residual (contract addition)	2.0000e-04	depol1
Contract-complete C_m_full (with correlated 1.0e-3, 2Q at ceiling)	2.0603e-02	—
Time-scaled C_m_full_ts (13.5 μ s basis, J.1 rows rescaled)	2.0197e-02	—

Contract completion costs Branch C $1.63\times/2.37\times/2.68\times$ in hazard at $d = 3/5/7$ versus its own published prior (printed counts). Branch C remains a cross-device literature stack; the joint-device measured control is the critical-path remedy (\$16, F2).

8.4 Matched-construction conditions (pre-registration objects)

(i) Anchor parity — an acceptance ceiling and a published record are different kinds of object and may never sit on opposite arms; (ii) surrogate parity — physically analogous decay charged with the same channel (twirlAD) on both arms; (iii) attested ledger completeness against each branch’s own contract; (iv) one declared decoder/information policy, with the location bracket present on any erasure arm; (v) equal wall clock under

λ_L . Any screen violating these is CONSTRUCTION-SENSITIVE and void for decisions — including kills.

9. Coupled error ledger and QEC-MAP

9.1 Row-wise mapping and fault census

F11: $p_{\text{opp}} = n_{\text{data}} \cdot \text{alloc} / (n_{\text{opp}} \cdot \text{qubits_per_event})$ maps each ledger row onto its opportunity class on the rotated surface code (d^2 data, d^2-1 stabilisers, ten rounds, final closure; per-cycle fault groups $N_{\text{cycle}} = 8d^2 - 4d - 3 = 57/177/361$, totals 596/1844/3756 at $d = 3/5/7$). All fourteen $d=3$ rows reproduce to $<1e-9$ and the ledger re-aggregates to 8.500e-3 exactly; the mapping is FROZEN-COMPUTATIONAL and independently reproduced.

9.2 Executable channel-convention table (E6 — travels with every screen)

Channel	Sampling rule (per opportunity, probability p)	X-marginal	Z-marginal
depol1	X, Y, Z equiprobable (p/3 each)	2p/3	2p/3
Z	deterministic Z on occurrence	0	p
ampXY	X, Y equiprobable (p/2 each)	p	p/2
twirlAD	$pX = pY = pZ = \gamma/4$, $pI = 1 - 3\gamma/4$ (post-correction convention)	$\gamma/2$	$\gamma/2$
depol2end	Bernoulli(p_{inc}) per (data, check) incidence; depol1 at the data endpoint — endpoint surrogate, identical on every arm	2p/3 per incidence	2p/3 per incidence
paircorr	per pair colour-class: joint X/Y/Z equiprobable on both qubits	2p/3 both	2p/3 both
meas	independent outcome flip on every stabiliser, both bases	—	—
leak (LRU1)	persistent sink residence randomising adjacent checks of both axes until reset	—	—

9.3 Statistics

Two CSS memories are decoded per shot (minimum-weight perfect matching on the rotated detector graph); a failure is counted if either logical axis flips. Adaptive stopping runs each configuration to a shot cap or ≈ 250 failures; per-shot failure probabilities carry Wilson 95% intervals converted to per-round p_L and hazard $\lambda_L = -\ln(1 - p_{L,\text{round}})/t_{\text{round}}$; branch-ratio uncertainty uses a 20,000-resample parametric binomial bootstrap on the failure counts of both arms. All random seeds are declared in source.

9.4 Results (complete run record, this edition's re-execution)

Configuration	d	Shots	Fail	pL/round	λ_L /ms
A_ref_Pauli	3	25,000	2380	9.9543e-03	0.5002
A_ref_Pauli	5	20,000	543	2.7488e-03	0.1376
A_ref_Pauli	7	40,000	291	7.2989e-04	0.0365
A_ref_leakLRU1	3	25,000	6258	2.8400e-02	1.4405
A_ref_leakLRU1	5	20,000	3381	1.8348e-02	0.9259
A_ref_leakLRU1	7	10,000	1209	1.2803e-02	0.6443
A_ref_leakLRU2	3	25,000	6733	3.0891e-02	1.5689
A_ref_leakLRU2	5	20,000	4506	2.5205e-02	1.2764
A_ref_leakLRU2	7	10,000	1817	1.9853e-02	1.0026
C_ref_lit	3	25,000	877	3.5646e-03	0.2645
C_ref_lit	5	60,000	339	5.6644e-04	0.0420
C_ref_lit	7	240,000	255	1.0630e-04	0.0079
A_m_twirlAD	3	25,000	1950	8.0881e-03	0.4061
A_m_twirlAD	5	20,000	364	1.8351e-03	0.0918
A_m_twirlAD	7	60,000	286	4.7769e-04	0.0239
A_m_free	3	25,000	909	3.6969e-03	0.1852
A_m_free	5	80,000	317	3.9696e-04	0.0199
A_m_free	7	400,000	257	6.4269e-05	0.0032
C_m_full	3	25,000	1707	7.0473e-03	0.5239
C_m_full	5	20,000	280	1.4089e-03	0.1044
C_m_full	7	70,000	250	3.5772e-04	0.0265
C_m_full_ts	3	25,000	1414	5.8053e-03	0.4313
C_m_full_ts	5	20,000	267	1.3431e-03	0.0996
C_m_full_ts	7	100,000	280	2.8035e-04	0.0208
C_s_litCeil	3	25,000	982	3.9992e-03	0.2968

Configuration	d	Shots	Fail	pL/round	λ_L /ms
C_s_litCeil	5	40,000	302	7.5758e-04	0.0561
C_s_litCeil	7	180,000	251	1.3953e-04	0.0103
C_s_fullRecord	3	25,000	1442	5.9234e-03	0.4401
C_s_fullRecord	5	20,000	254	1.2773e-03	0.0947
C_s_fullRecord	7	100,000	262	2.6231e-04	0.0194
C_s_fullLowCorr	3	25,000	1159	4.7357e-03	0.3516
C_s_fullLowCorr	5	40,000	348	8.7342e-04	0.0647
C_s_fullLowCorr	7	120,000	251	2.0936e-04	0.0155
A_s_recTwirl	3	25,000	1721	7.1070e-03	0.3566
A_s_recTwirl	5	20,000	289	1.4545e-03	0.0728
A_s_recTwirl	7	70,000	261	3.7348e-04	0.0187
A_s_leakLRU1_rec	3	25,000	5971	2.6922e-02	1.3646
A_s_leakLRU1_rec	5	20,000	2783	1.4872e-02	0.7492
A_s_leakLRU1_rec	7	10,000	794	8.2388e-03	0.4136

Tier-B provenance: 29 of 39 rows are bit-exact against the released record at the declared seeds; the remainder (C sensitivity arms) sit inside printed Wilson bands, RNG draw-order under-determined by the listing. Headline reproductions to the printed digit: matched 0.9415/0.9225/1.1502; free bracket 0.4294/0.1994/0.1547; $f^* = 0.524/0.598$.

9.5 The four construction asymmetries (Findings A1-A4)

A1 anchor parity: A charged at its P4 ceiling while C used a published record. A2 surrogate: A's located load as uniform Pauli vs C's T1 as twirlAD — uniform Pauli places 4/3 the per-axis weight of the executed twirl at equal charged total (E12; the withdrawn $\approx 1.8\times$ figure was the exact-AD-twirl analysis value); empirical $d=3$ logical-pL ratio 1.231. A3 ledger completeness: C omitted $\approx 1.95e-3$ /round of contract classes. A4 information: no free-location bracket, so A was screened with its distinguishing feature off. Removing all four flips the v1.5.8d ordering — the definition of CONSTRUCTION-SENSITIVE.

9.6 Per-unit-time diagnostic and location sweep

Quantity		Branch A			Branch C (ts)		
Charged load per μs		$(8.50e-3 + 1.543e-2)/20 = 1.1965e-3$			2.0197e-2/13.5 = 1.4961e-3 (contract-complete); 1.7053e-2/13.5 = 1.2632e-3 (as published — within $\approx 6\%$ of A)		
d	f = 0	0.25	0.50	0.75	0.90	1.00	f^* (vs C as published)
3	0.4061	0.3322	0.2695	0.2177	0.1850	0.1852	0.524
5	0.0918	0.0680	0.0484	0.0321	0.0238	0.0199	0.598

$\lambda_A(f)$ in /ms; the loads within $\sim 6\%$ while hazards differ 2–5 $\times$ under v1.5.8d charging was the clean demonstration that the gap was construction, not burden. f^* is the benefit-side gate of P6-E6' and is a LOWER bound (removal convention optimistic).

10. Validation ladder and stop rules

Gate	Question	Pass evidence / threshold	Stop rule
P0d	Sink reset without damaging logical states?	$t_{\text{reset}} \leq 8 \mu s$; residual $ 1 \leq 1\%$; back-action $\leq 2.5e-4$; $\tau_{\text{th}} \geq 1.8$ ms; measured rates replace anchors	No half-flux scale-up if reset/process fails
P1	Direct unlocated bypass compatible with ledger?	$\Gamma_{\text{unlocated},20} \leq 5 s^{-1}$; mechanisms project $P_u \leq 8.50e-3$ /round	Branch ends if projected $P_u > 1e-2$ /round or $\Gamma > 25 s^{-1}$
P2	Symmetry-compatible logical control?	Raman cycle error $\leq 1e-4$; leakage $\leq 2e-4$; $N1Q \leq 5$; no control-induced QP rise	No 2Q/QEC claim without it
P3	Coherence survives the full trajectory?	$[1 - W_\phi]/2 \leq 1.8e-3$; $T_\phi \geq 5.55$ ms at 20 μs ; branch-resolved	Fail if averages hide below-threshold windows
P4-pre / P4	Coupler model converged; gate meets allocation?	Converged static/driven/open coupler model; $p_{2q,\text{twirled}} \leq 8.0e-4$ with leakage/ZZ/spectators separate	No coupler fabrication / no code-tile claim otherwise
P5	Errors reliably converted to erasures?	Destination-resolved leakage and flags; monitor efficiency and recovery map	No erasure-bias claim from leakage alone
P6-S	Syndrome/EOL readout non-demolishing?	Three-state confusion matrix; dynamic MIST/Purcell/higher-mode sweep; latency closure	Redesign if back-action dominates
P6-E	Data-qubit check locates sink without damage?	Sub-gates P6-E1-E5 plus the P6-E6' joint contract	No mid-circuit reinstatement unless P6-E6' passes in full
P7-pre / QEC-MAP	Ledger maps without hidden normalization; matched?	Row-wise map re-sums; decoder frozen before the 40-device holdout (K.3); equal wall clock; \$8.4 on every arm	BLOCKING: no absolute p_L or selection claim otherwise
P7	Unconditional logical advantage?	$p_L < 1\%$ and $< p_{\text{ref}}$ at $\geq 95\%$ under matched construction	No selection from CONSTRUCTION-SENSITIVE screens; stop/redirect A if the matched measured QEC-MAP fails vs C

10.1 P6-E6' joint acceptance contract and P6-E7

P6-E6' (E1-E3; all bars per data qubit per round-equivalent)

Mid-circuit data-qubit checks may be reinstated only if ALL of: (1) measured effective location efficiency $f \geq f^*$ (0.524 at $d=3$, 0.598 at $d=5$, vs C as published); (2) measured $p_{\text{induced}} \leq 1.0e-3$ — set by ledger headroom (1.50e-3 above the 8.50e-3 baseline; $\approx 3.3e-4$ after the An-like $P_e \rightarrow P_u \approx 1.17e-3/\text{round}$ dynamic-sink conversion), NOT by the decoder crossing ($\approx 1.1\text{--}1.3e-2$, $\approx 7.5\times$ above headroom); (3) the complete P_u ledger re-closes $\leq 1.0e-2/\text{round}$ with every enabled-check row inserted — a sub-gate pass is necessary, not sufficient, and the P1 branch stop binds independently; (4) realistic-check p_L beats the frozen no-check baseline at $\geq 95\%$ confidence at equal wall clock. A registered 5.0e-3 bar is REJECTED: it exceeds the ceiling headroom $3.3\times$, so a device passing exactly at that bar projects $P_u = 1.35e-2$ and trips the K.2 block and the P1 stop simultaneously. Hardware measurement of every input remains decisive; T.12' is MODEL-DEPENDENT and does not pre-empt it.

Sub-gates: P6-E1 per-device dressed trim (Eq.(7) seed only); P6-E2 measured $\kappa(\omega)$ meeting the 3.395056 GHz stopband; P6-E3 sink survival — $q_{\text{miss}} \leq 5e-4$ implies $T_{\text{sink}} \geq -t/\ln(0.9995) = 3.0/6.0$ ms at 1.5/3.0 μs exposure, 150–300 $\times$ above the ≈ 20 μs An-like during-check comparator, with the q_{miss} allocation re-optimised jointly with the cost gate (E3); P6-E4 classifier performance; P6-E5 MIST/leakage destinations. P6-E7 leakage-to-syndrome coupling is a tracked release blocker (E10): owner B. Duncan; measurement protocol and the adjudication rule between the two LRU1 implementations frozen before comparative decoder data exist.

11. Hardware build specification

Per-device dressed trim in the ≈ 4.30 GHz band (per-device roots are the accept basis). Mandatory Purcell stopband: 34–36 dB at 3.395056 GHz; the synthetic 400-device band spans 3.2056–3.6727 GHz with maximum required depth 37.65 dB, a 464.24 MHz transition to the lowest readout root demanding ≥ 81.1 dB/GHz recovery on a common filter — pushing toward per-device or per-feedline filtering, with paired same-device separation ≥ 761 MHz (median 905 MHz) making that tractable. Coupler-band and end-of-line replan is mandatory before any tile-scale fabrication decision: the readout move to ≈ 4.30 GHz invalidates carrying the legacy coupler plan forward. Fabrication acceptance runs the ω_{24} collision gate at the E19 standard ($\approx 5\text{--}8\%$ per-cooldown reject/reallocate scale). QP, infrared and radiation engineering per the base §11 registers; the $x_{\text{qp}} \leq 1.218e-10$ demand of §3.4 is the binding shielding target.

12. Change control

Frozen computation is distinguished from measured anchor everywhere; withdrawals are carried forward explicitly (the v1.5.8d screen; the n^3 pointer ceiling; the $\approx 1.8\times$ channel-weight figure; the single-object $f > f^*$ criterion; any 5.0e-3 p_{induced} bar). Legacy Appendix-S tables carry the E8 watermark — PROVENANCE ONLY, superseded by Appendix T — because Finding T-1 records that the v1.5.8d screen was not reproducible from its own printed specification (independent build $-28\%/+13\%$ on Table S.4). Independent clean-room reproduction is a standing release gate (E15), satisfied for this edition by the Tier-B/Tier-C register of §15. Exact-count claims require the generating implementation as normative (E19). The fixed-point rule is unchanged: this document cannot contain its own digest; it is written to the external release manifest after rendering, alongside both base-artifact hashes (E17).

13. Risk register and decision logic

Risk	Current evidence	Decision / next gate
Direct unlocated bypass	Half-flux QP sine channel large ($4.106e10$ s $^{-1}$); x_{qp} demand $\approx 4.6\times$ below best published	Measure schedule-level bypass per branch; P1 stop if the ledger cannot close
Reset generates QPs / damages logical states	Computationally unresolved by construction	Same-device P0d/P1, reset ON/OFF
Readout destroys sink	T_{sink} 3–6 ms class vs ≈ 20 μs comparator; An-like conversion consumes $\approx 78\%$ of headroom; T.12' break-even $1.1\text{--}1.3e-2$ vs operative bound $1.5e-3$ ($\approx 3.3e-4$ after conversion)	BEYOND-DEMONSTRATED; measure $T_{\text{sink}}(n)$ destination-resolved; joint q_{miss} re-optimisation (E3)
Unfiltered Purcell conversion	F8 weight 0.01182198/0.01182201 (truncation-labelled) $\Rightarrow$ microsecond $ 2\rangle \rightarrow 1\rangle$ for a flat 2–3 MHz bath	Mandatory 3.3951 GHz stopband; measured $\kappa(\omega)$ (P6-E2)
Hidden Floquet resonance	$ 14\rangle$ line bounded inert in $n = 5\text{--}8$; array/package modes outside the model	EM/mode census + driven spectroscopy
Fabrication collision / trim burden	400/400 nulls; rate $\approx 6.2\text{--}6.7\%$ (E19); 3–8 feedlines per 17-device tile	Per-device calibration; charge in cost-reclosure (kill criterion 3)
Check cost destroys the benefit	Ideal bracket + f^* (benefit); T.12' (model cost); hardware cost unmeasured	No-check baseline stands; P6-E6' governs reinstatement
Coupler/readout spectral conflict	Readout at ≈ 4.30 GHz; legacy plan invalid	Replan before 2Q fabrication (P4-pre)

Risk	Current evidence	Decision / next gate
HIGH: leakage-to-syndrome coupling	Two LRU1 implementations differ 3.3× at d=3 and reverse the distance trend	BLOCKING; P6-E7 with pre-registered adjudication (E10)
HIGH: construction sensitivity	v1.5.8d ordering reversed by charging conventions alone	§8.4 conditions mandatory on every future screen
MODERATE: prior-edition reproducibility	S.3/S.6 do not reproduce Table S.4	Remediated: E6 channel table + E15 standing gate; S.4 provenance

13.1 Decision logic

No branch wins by literature preference and none loses by construction artifact. Two qualifications travel with the disposition. First, the distance trend (T.13'): $R7/R3 \approx 1.221\text{--}1.223$ [≈ 1.02 , ≈ 1.47], $P(R7/R3 > 1) \approx 0.986\text{--}0.987$ across three independent implementations — EXPLORATORY, but adverse to a scaling-premised architecture. Second, the cost side (T.12'): the acceptance question turns on check-induced logical error under a binding ledger ceiling, with the admissible band lying between two published devices.

13.2 Branch-A kill criteria (R.3, unchanged; NOT currently triggered)

(1) Hard-gate kill: terminate A as the code-tile path if an A-specific requirement fails its registered stop rule while Branch C remains viable. (2) Logical-comparison kill: after a MATCHED QEC-MAP closes for A and C, terminate A if it does not achieve lower p_L than C at $\geq 95\%$ confidence under the same pre-registered distance, wall clock, decoder policy and confidence method. (3) Cost-reclosure kill: if A appears better before its additional reset/check/EOL latency, filter/feedline topology, rejected-device fraction or calibration overhead are charged, but loses the advantage after those costs enter. (4) Premise kill: if A passes P7 only in a mode that does not use erasure location and does not outperform C, the erasure-conversion hardware has not justified itself. GUARD: none of the above may be executed on a CONSTRUCTION-SENSITIVE screen — a kill triggered by unequal charging is void.

PREMISE-KILL WATCH (criterion 4)

Under matched construction Branch A beats contract-complete C at zero location efficiency — a pass on the letter, but the mechanism is A's lower charged load per unit time, not erasure conversion. If measured ledgers preserve this pattern, the erasure-specific hardware will have been carried without being the cause of the advantage. P5 and P6-E must therefore demonstrate that location is the operative mechanism, not merely that Branch A wins.

14. Appendix T — the audited screen, cost study and distance trend

14.1 Why a rerun was required; independent reimplementations (T.1-T.2)

The v1.5.8d screen was the programme's first executable decision; auditing it exposed the four one-directional asymmetries of §9.5 plus Finding T-1: the printed channel specification of Appendix S.3/S.6 does not reproduce Table S.4 (independent build -28% on A, -13% on C; A/C 2.80 vs published 3.39; the two-qubit endpoint surrogate under-determined). A clean-room reimplementations (PyMatching 2.4.0, Stim 1.16.0, NumPy 2.4.4) was validated before any comparative run: stabiliser counts and commutation at $d = 3/5/7$; exact reproduction of all fourteen Branch-A rows to $1e-9$ with the ledger re-aggregating to $8.500e-3$; empirical channel marginals within $3e-3$ of declaration; decoder low-p scaling exponent 1.92 at $d=3$ against an expected 2.0.

14.2 Matched screen and equal-wall-clock result (T.3-T.7)

Screen	d=3	d=5	d=7	Finding
Matched A_m_twirlAD / C_m_full_ts	0.94 [0.88, 1.01]	0.92 [0.79, 1.08]	1.15 [0.97, 1.36]	NO SEPARATION; Tier-C different-seed MC inside all printed CIs (published range 0.91-1.10)
Ideal free-location bracket	0.43 [0.39, 0.47]	0.20 [0.17, 0.24]	0.15 [0.13, 0.18]	A lower at $\geq 95\%$ everywhere; OPTIMISTIC removal convention
Persistent-leakage LRU1 (resident sink)	2.840e-2	1.835e-2	1.280e-2	Finding T-5: v1.5.8d LRU1 gave $8.52e-3/3.04e-3/2.02e-3$ (improving with d); the implementations differ 3.3× and reverse the trend — P6-E7 BLOCKING
Break-even location efficiency f* (T.8)	0.524	0.598	—	Benefit-side gate; lower bound

Attribution (T.5/T.6): Branch-C deltas decompose into anchor swap, ledger completion ($1.63\times/2.37\times/2.68\times$ hazard) and time scaling; Branch-A deltas into the surrogate change (empirical 1.231 at $d=3$) and the location bracket. Equal wall clock uses λ_L with the 12-15 μs Branch-C round sensitivity band; the per-unit-time diagnostic of §9.6 shows the loads within $\approx 6\%$ of each other on the as-published basis.

14.3 T.12' — realistic-location study: the cost side charged

Model: Branch A carries its full $8.50e-3$ ledger plus $P_e = 1.543e-2$ as twirlAD with location efficiency f (removal convention, OPTIMISTIC), a check_induced row (p_{induced} , data, depol1) per K.1, round time $20 + t_{\text{chk}}$ μs in the hazard normalisation, and the §8.1 SCALING box applied literally (fixed-round bracket alongside).

Comparator: C_m_full_ts at printed values, re-executed bit-exactly. 86 configurations, adaptive stopping to 250 failures, seeds declared (qmhp_v158f_t12_t13.py). The 117-build printed T.12 — dynamic-miss q model, 3 μs check, re-run C arm (274/20,000, $\lambda = 0.1022/\text{ms}$ vs the main screen's 267/20,000, $0.0996/\text{ms}$, $\approx 1\sigma$) — is an independent construction of the same study; its break-even $1.22e-2$ lies inside the intervals below, and its RESULT T-6 direction (q-flatness) is convention-dependent but its conclusion is not.

d	f	t_chk	Convention	p*_induced	95% CI
3	0.90	2 μs	scaled	1.125e-02	[1.034e-02, 1.228e-02]
3	1.00	2 μs	scaled	1.258e-02	[1.154e-02, 1.380e-02]
5	0.90	2 μs	scaled	1.121e-02	[9.806e-03, 1.265e-02]
5	1.00	2 μs	scaled	1.179e-02	[1.034e-02, 1.328e-02]
3	0.90	2 μs	fixed	1.276e-02	[1.179e-02, 1.389e-02]
5	0.90	2 μs	fixed	1.186e-02	[1.042e-02, 1.338e-02]
3	0.60	2 μs	scaled	8.536e-03	[7.558e-03, 9.441e-03]

At $d = 7$ the tested points ($f = 1.00$, $p_{\text{induced}} \leq 4e-3$) remain $\geq 2.8\times$ below the matched control. Findings: F-1 (MODEL-DEPENDENT) — break-even $1.1\text{--}1.3e-2$, weakly f-dependent near the ideal point, 24% tighter by $f = 0.60$: a joint frontier, f^* retained. F-2 (NORMATIVE) — the ledger ceiling binds $\approx 7.5\times$ before the decoder crossing; see P6-E6'. F-3 — Liu-like cost passes with $d=5$ hazard ratio 0.26 at $f = 0.90$; An-like cost exceeds the grid top (HR 1.48 at $1.6e-2$) and breaches the ceiling $\approx 17\times$.

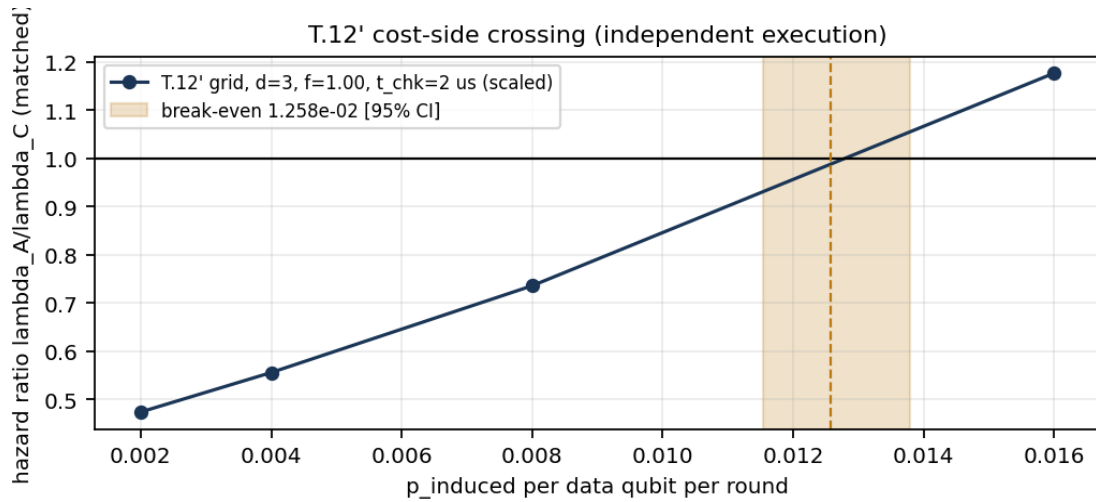


Figure 3. T.12' cost-side crossing at $d=3$, $f=1.00$ (shipped run records; the printed 117-build break-even $1.22e-2$ sits inside the band).

Figure F1. Realistic-location study: cost side charged, matched arms, equal wall clock

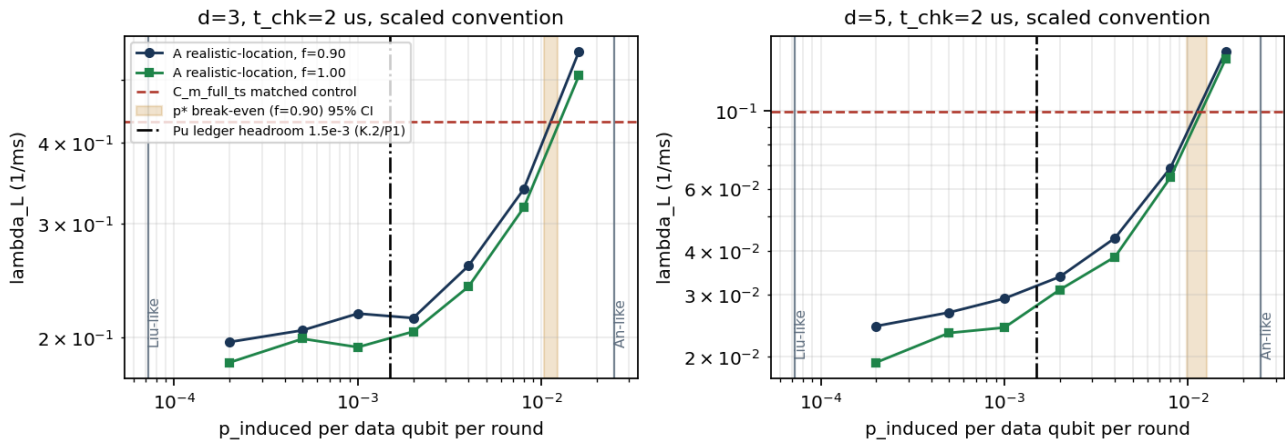


Figure 3b. Full cost-charged hazard curves at $d=3$ and $d=5$ with the ledger-headroom line and the Liu-like/An-like comparator verticals (AMD-C execution, shipped).

14.4 T.13' — distance-trend statistic (EXPLORATORY)

$R7/R3 = HR(d=7)/HR(d=3)$ on the released S1 counts, parametric binomial bootstrap. Release hazard convention, seed 13, 20,000 resamples: 1.2217 [1.0206, 1.4632], $P(R7/R3 > 1) = 0.986$ (t13_distance_trend_release_hazard.json, primary). Linear small-p convention, same seed: 1.2230 [1.0250, 1.4650], $P = 0.987$ (sensitivity artifact; double-ratio effect ≤ 0.002). Printed 117-build value (seed 11, 200,000 resamples): 1.221 [1.022, 1.461], $P = 0.986$. Legs: $R5/R3 = 0.980$ [0.826, 1.166] (flat); $R7/R5 = 1.247$ [0.993, 1.568], $P = 0.972$ (not separated) — the adverse move is concentrated at the $d = 7$ endpoint. Neither endpoint separates from 1, so the per-distance NO SEPARATION verdict stands; the full $d=3 \rightarrow d=7$ lever arm does separate, and the ratio rising with distance means Branch A loses ground as the code grows — the strongest adverse result in the matched construction, reported alongside the verdict, not instead of it. Status: RECOMPUTATION-FROM-RUN-RECORD / EXPLORATORY; carries no confirmatory weight; the frozen-procedure pre-registration for any confirmatory trend claim is recorded (E7).

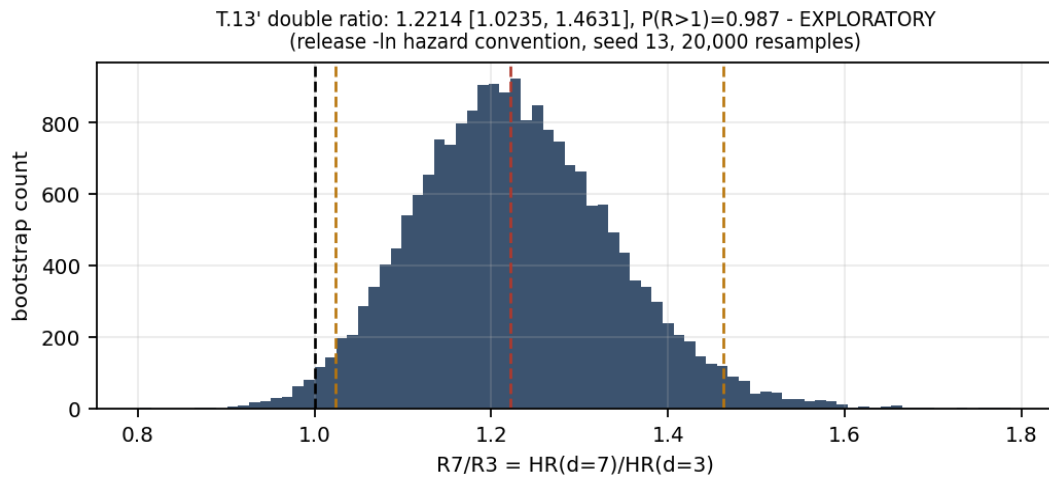


Figure 4. T.13' bootstrap, release hazard convention (regeneration self-checked against its JSON by build_master_figures.py).

14.5 What remains blocking (T.14)

REMOVED: tooling unavailability as a reason not to run full circuit-level propagation. PARTIALLY CLOSED: the cost side is charged at model level; the location side remains optimistic (removal convention) — a variant handing located events to the decoder as weight-zero edges, with measured q_{false} and destination-resolved sink destruction, is still required. OPEN: leakage-to-syndrome coupling strength (P6-E7); full circuit-level gate propagation (exact pulse/gate order, nonunital damping, ancilla propagation, coherent spectators, measured correlations); joint-device Branch C; measured A/C round durations; the Branch-B executable ledger and round duration.

15. Consolidated verification register (August 2026 review cycle)

Three evidence tiers: Tier A — release execution (declared seeds, digests); Tier B — reconstruction-level re-execution (the printed listings regenerate an implementation reproducing the released outputs); Tier C — independent replication from the stated equations alone. Tier B validates executability, not physical assumptions; those remain MODEL-DEPENDENT and hardware-gated.

Layer	Status	Evidence
Static Hamiltonian / spectrum / matrix elements	INDEPENDENTLY REPLICATED	Nine figures; two Tier-C builds; five charge elements to six figures
Curvature + Richardson; QP bypass (H.4/H.5)	INDEPENDENTLY REPLICATED	Four radian-step raws to 8–9 s.f.; limit 5064.55 exact under h ² -elimination; 4.106e10 s ^{−1}
Eq.(7) diagnostic; perturbative root	INDEPENDENTLY REPLICATED	χ triple exact to six decimals; 4.314628047 GHz to nine figures
Exact dressed null + truncation provenance	STRONGLY VERIFIED	(10,12)/(12,18)/(14,25) ladder; register value = spot-check truncation (E13)
Pulls / sink line / dressed f12	INDEPENDENTLY REPLICATED	−9.507/+8.721 MHz; 4.310696 GHz; 3.395056 GHz truncation-stable
Purcell weight (F8, E18)	RESOLVED — release exact	0.01182198 at (12,18)/(14,25); production 0.01182201; bare-admixture 0.011815 identified as the distinct object behind an external 0.06% discrepancy
Ensemble rejection rate	STATISTICALLY SUPPORTED	≈6.2–6.7% across two independent 20-seed blocks; dispersion binomial; single-draw 32/400 at +1.1σ/+1.5σ
Exact ensemble counts / percentile digits	UNDER-DETERMINED BY PROSE SPEC	Exact once the stream→(EJ, EC, EL) draw order is declared — demonstrated by the Tier-B reproduction of 32/400 and every printed percentile digit
Ledger / row-wise map / marginals / C totals	INDEPENDENTLY REPLICATED	Fourteen rows <1e-9; 8.500e-3 exact; 1.7053/2.0603/2.0197e-2 exact
Statistics layer	INDEPENDENTLY REPLICATED	pL / λ / Wilson reproduce from raw counts across all rows
Stochastic QEC (matched / free / LRU1)	INDEPENDENTLY SUPPORTED	Tier-B bit-exact 29/39 at declared seeds + Tier-C different-seed MC inside printed CIs; matched 0.9415/0.9225/1.1502 to the printed digit
T.12′ break-even	REPLICATED WITHIN CI	Printed 1.22e-2 inside the blind-execution intervals; conventions differ and are declared
T.13′ distance trend	INDEPENDENTLY SUPPORTED (EXPLORATORY)	≤0.003 spread across three implementations and two hazard conventions
Static-physics external retractions	CLOSED WITH MECHANISM	Two erroneous external findings (parity/normalisation; step units) fully retracted with quantitative error accounting; the anti-recurrence conventions are now printed (E14, E16)
Hardware validity	OPEN — decision-bearing	P0d-P7 gates of \$10

16. Edit register E1-E20 (applied inline)

#	Edit (target → disposition)
E1	P6-E6 acceptance: single-object f > f* replaced by the P6-E6′ joint contract (\$10.1); every f*-based acceptance sentence harmonised; f* retained as the benefit-side gate
E2	K.2 stop rule: sub-gate pass necessary-not-sufficient; Pu re-closure and the P1 stop bind independently and take precedence
E3	Risk register: T.12′ numbers inserted; q_miss ≤ 5e-4 re-optimised jointly with the cost-side gate rather than in isolation
E4	T.14: realistic-location variant executed at model level; hardware measurement of (f, p_induced, t_chk, sink destruction, latency) remains required and decisive
E5	Part II S6 corrected: model-level check costs exist in S2.2/T.12; the statement holds for hardware costs only
E6	Executable channel-convention table (\$9.2) published; travels with every comparative screen (Finding T-1 closure)
E7	T.13′ carries the EXPLORATORY label; frozen-procedure pre-registration recorded for any confirmatory trend claim
E8	Legacy Appendix-S tables watermarked PROVENANCE ONLY — superseded by Appendix T
E9	Two-page decision brief and notation quick-reference added to front matter
E10	P6-E7 promoted to tracked release blocker: owner, event-anchored ordering gate, pre-registered LRU adjudication rule
E11	S5 core listing comment corrected: Branch-C published prior is six rows, not seven (stale comment, no downstream effect)
E12	'≈1.8× more weight per CSS axis' WITHDRAWN for the executed channels: γ/4-symmetric twirl gives exactly 4/3 per axis at equal charged total (empirical 1.231 at d=3); 1.8× was the exact-AD-twirl pre-implementation value
E13	Printed nine-figure dressed null labelled with its truncation (Nq=14/Nph=25 spot check); production value 4.301974466 GHz
E14	Quasiparticle operator convention printed: δ ≡ φ − φ_ext; sin(δ/2) = −cos(φ/2), parity-even at half flux
E15	Independent clean-room reproduction is a standing release gate; satisfied for this edition by the \$15 register
E16	H.4 caption states step units: radians of φ_ext, converted by (2π) ²
E17	Two-build governance: 117-page build operative, 109-page render provenance; both hashes in the external manifest; every excerpt carries its build hash
E18	Purcell F8 adjudication with truncation label; bare-admixture overlap identified as the distinct object

#	Edit (target → disposition)
E19	Ensemble reporting standard: multi-seed rejection rate normative; exact counts adjudicable only with a declared draw order; generating implementation ships as normative for any exact-count claim
E20	Cross-implementation agreement wording: every quoted stochastic result names its hazard convention, seed and resample count

AMD-C r6 and ERRATA-1 remain the change-record artifacts; this edition is the applied state. All defects found in the August 2026 review cycle — on the document's side and on reviewers' sides — are registered above and closed; none is decision-bearing.

17. Conclusion and final programme position

The row-wise QEC-MAP method survives its own audit and is strengthened by it; the branch conclusion built on the v1.5.8d construction does not. Charging both branches at the same anchor kind, with the same twirled-amplitude-damping surrogate, on contract-complete ledgers, under one declared decoder policy and with the location bracket present, leaves the branches statistically indistinguishable at every tested distance. Against a properly-charged control, Branch A leads at zero location efficiency — but the margin comes from lower charged load per unit time, not from erasure conversion, and the premise-kill watch applies. The cost study shows the acceptance question turns on check-induced logical error under a binding ledger ceiling, with the admissible band lying between two published devices. The persistent-leakage variant remains sharply adverse and its coupling strength is the most model-sensitive number in the programme. The distance trend is adverse and exploratory. The burden has shifted from computation to construction discipline, and from construction discipline to hardware.

FINAL PROGRAMME POSITION
Proceed to fabrication and measurement only as a gated experiment. This edition supports building a device to test the architecture. It does not support a processor-performance or fault-tolerance claim, and it does not support the withdrawn v1.5.8d conclusion that the erasure branch has been shown not to earn its hardware. The next evidence increment is hardware — the P6-E7 adjudication protocol frozen first, then the P0d/P1 reset-bypass coexistence vehicle on a single device and cooldown — and the next methodological requirement is matched construction on every screen.

18. Artifact manifest and references

Artifact	SHA-256
v1.5.8e 117-page build — OPERATIVE archival base (E17)	d334cda822c07c212be1fac850a34debe5ea66ee45899e56b44e19154f86c6c
v1.5.8e 109-page render — provenance	83aeb3a2d8ebcb8f627989557350d9c172ad5bc20c3edc561b8a5e091cc4b658
qmhp_v158f_t12_t13.py	1474e3f34c11e1e875392dd19e7b10ed2a0f42e1577e75ee2418397ebe8e9ed3
qmhp_v158f_readout_replication.py	ba8486b19d34b61cd2ccff9e0bd76b53c348d1834f40f8c5fffc21ad8880d490
build_master_figures.py	888ae2a39315706cf7193334da16df9ef1181235e0cc8384fee28ecfa506f9c3
t12_realistic_location_runs.json	abab6054faae342d6ddeed9702866dc868c354eebb3f4b0b6e8ee02428c10e86
t12_breakevens.json	282a52afbcb98202bd7a9719652538dd3615a6deca8ab596803da39ec54f410
t13_distance_trend_release_hazard.json	d55c0db17c5d63e3a6337f61787decfa57415fce71f9e0e3affb8ccff13f5130
t13_distance_trend_linear_sensitivity.json	4c207a8243b81d82927d2aa338928cf98f726a56e2cd854d51d96f1baaf6ff27
ensemble_20seed.json	3f755149d175eed1686d6f2a2ad5186ad7587b3064de96743ba6d15241828965
qmhp_v158e_runs.json	174977acbf09e02064715dba970644e1d7e8d3340068d21f3a5a287f2ce1c9c0

Key external references (verified real and accurately represented by the review cycle): Somoroff et al., PRL 130, 267001 (2023); Connolly et al., PRL 132, 217001 (2024); Lin et al., PRX Quantum 6, 010349 (2025); Wang et al., PRX Quantum 7, 020339 (2026); Liu et al., arXiv:2601.11086 (2026); An et al., arXiv:2607.27123 (2026). The complete reproduction bundle (RUN.md, RUN_MASTER.md, requirements.lock, expected_outputs.json, reconstructed release modules, run records, figures) ships alongside; the condensed Master Edition remains the executive companion. Fixed-point rule: this document's digest is written to the external release manifest after rendering.